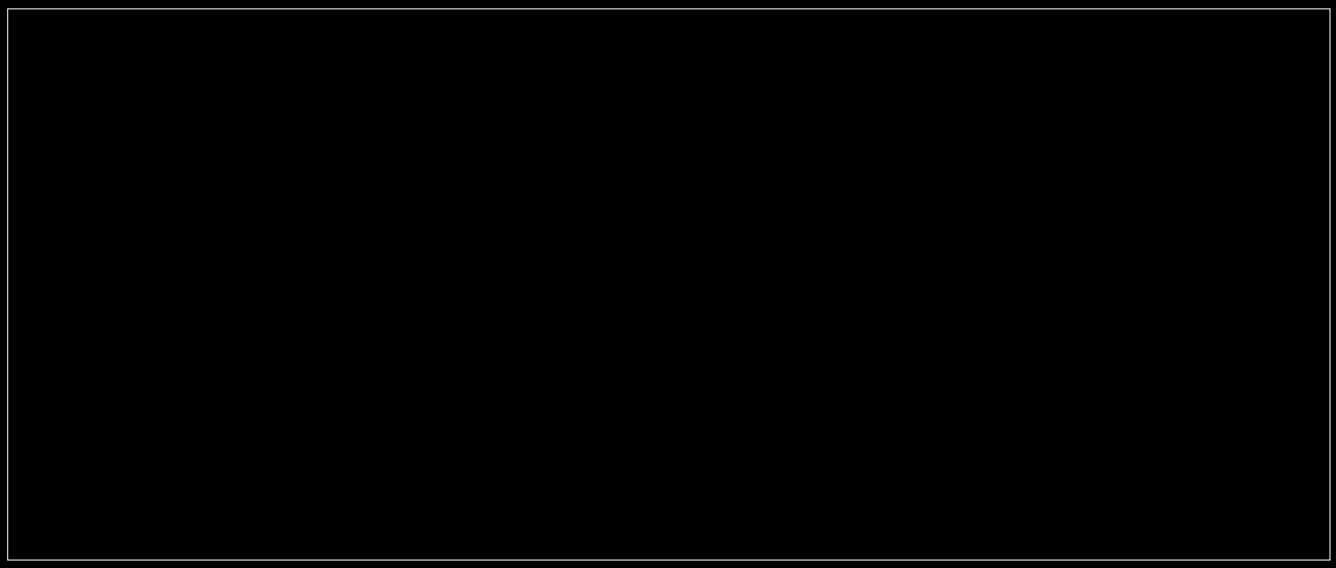
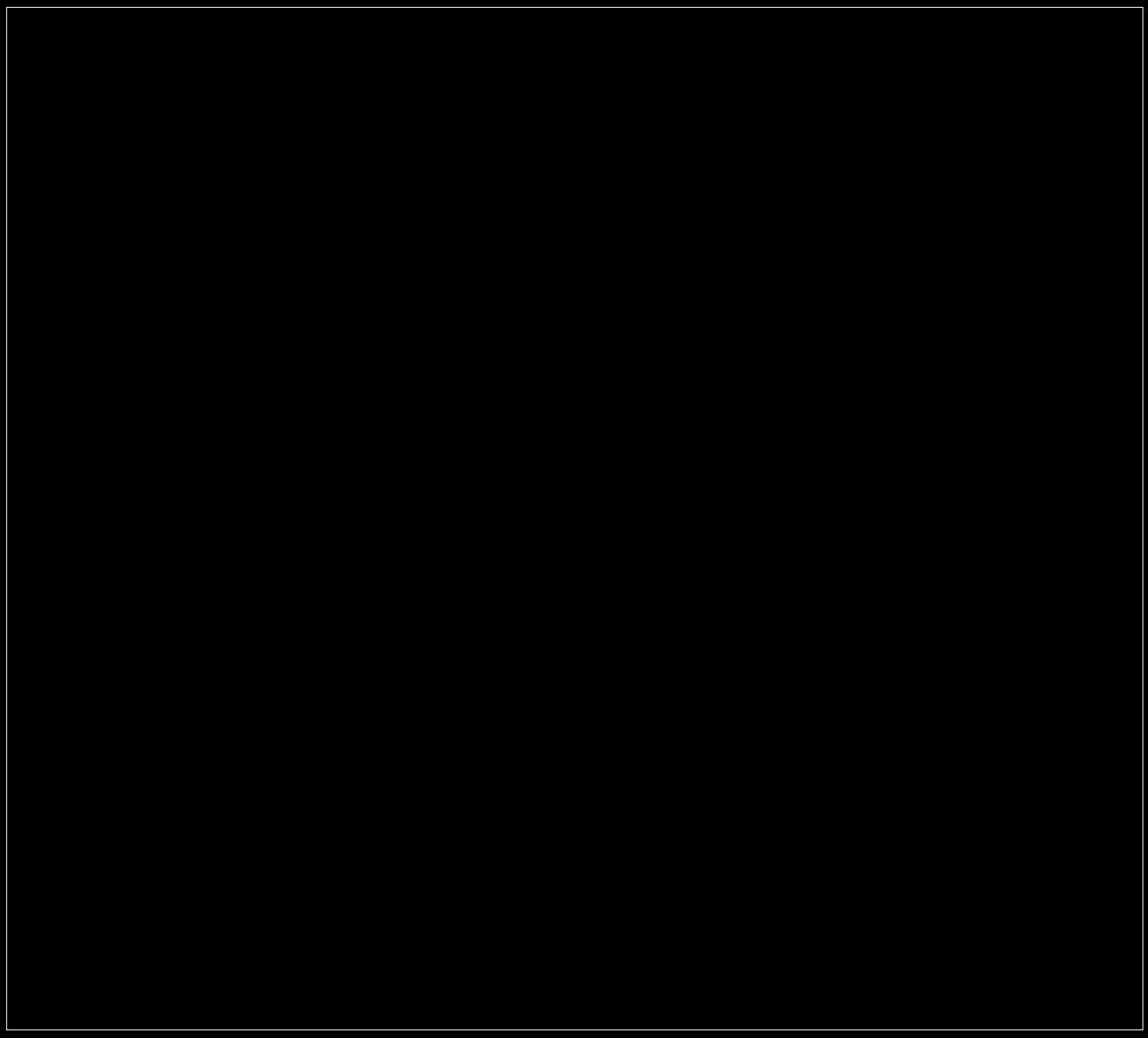




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UNIVERSITY *of* CAMBRIDGE
International Examinations

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Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1.$$

$$\sec^2 A = 1 + \tan^2 A.$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A.$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

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1 Express each of the following statements in appropriate set notation.

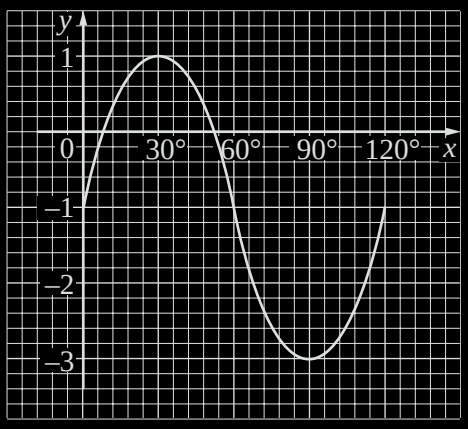
(i) x is not an element of set A.

(ii) The number of elements not in set B is 16.

(iii) Sets C and D have no common element.

[3]

2



The diagram shows part of the graph of $y = a\sin(bx) + c$. State the value of

(i) a,

(ii) b,

(iii) c.

[3]

3 The equation of a curve is $y = \dfrac{8}{(3x-4)^2}$.

(i) Find the gradient of the curve where $x = 2$. [3]

(ii) Find the approximate change in y when x increases from 2 to $2 + p$, where p is small. [2]

4 The vector $\overrightarrow{OP}$ has a magnitude of 10 units and is parallel to the vector $3\mathbf{i} - 4\mathbf{j}$. The vector $\overrightarrow{OQ}$ has a magnitude of 15 units and is parallel to the vector $4\mathbf{i} + 3\mathbf{j}$.

(i) Express $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ in terms of **i** and **j**. [3]

(ii) Given that the magnitude of $\overrightarrow{PQ}$ is $\lambda\sqrt{13}$, find the value of λ. [3]

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5 A large airline has a fleet of aircraft consisting of 5 aircraft of type *A*, 8 of type *B*, 4 of type *C* and 10 of type *D*. The aircraft have 3 classes of seat known as Economy, Business and First. The table below shows the number of these seats in each of the 4 types of aircraft.

Type of aircraft \ Class of seat	Economy	Business	First
A	300	60	40
B	150	50	20
C	120	40	0
D	100	0	0

(i) Write down two matrices whose product shows the total number of seats in each class.

(ii) Evaluate this product of matrices.

On a particular day, each aircraft made one flight. 5% of the Economy seats were empty, 10% of the Business seats were empty and 20% of the First seats were empty.

(iii) Write down a matrix whose product with the matrix found in part **(ii)** will give the total number of empty seats on that day.

(iv) Evaluate this total.

[6]

6 Given that the coefficient of x^2 in the expansion of $(k + x)\left(2 - \frac{x}{2}\right)^6$ is 84, find the value of the constant k.

[6]

7 The function f is defined for the domain $-3 \leqslant x \leqslant 3$ by

$$\mathrm{f}(x) = 9\left(x - \frac{1}{3}\right)^2 - 11.$$

(i) Find the range of f. [3]

(ii) State the coordinates and nature of the turning point of

(a) the curve $y = \mathrm{f}(x)$,

(b) the curve $y = \left|\, \mathrm{f}(x) \,\right|$.

[4]

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8 **(a)** Solve the equation $\lg(x + 12) = 1 + \lg(2 - x)$. [3]

(b) Given that $\log_2 p = a$, $\log_8 q = b$ and $\frac{p}{q} = 2^c$, express c in terms of a and b. [4]

9 A curve has the equation $y = \frac{2x - 4}{x + 3}$.

(i) Obtain an expression for $\frac{dy}{dx}$ and hence explain why the curve has no turning points. [3]

The curve intersects the x-axis at the point P. The tangent to the curve at P meets the y-axis at the point Q.

(ii) Find the area of the triangle POQ, where O is the origin. [5]

10 The cubic polynomial $f(x)$ is such that the coefficient of x^3 is 1 and the roots of $f(x) = 0$ are 1, k and k^2. It is given that $f(x)$ has a remainder of 7 when divided by $x - 2$.

(i) Show that $k^3 - 2k^2 - 2k - 3 = 0$. [3]

(ii) Hence find a value for k and show that there are no other real values of k which satisfy this equation. [5]

11 **(a)** Solve, for $0° \leqslant x \leqslant 360°$, the equation

$$2\cot x = 1 + \tan x.$$ [5]

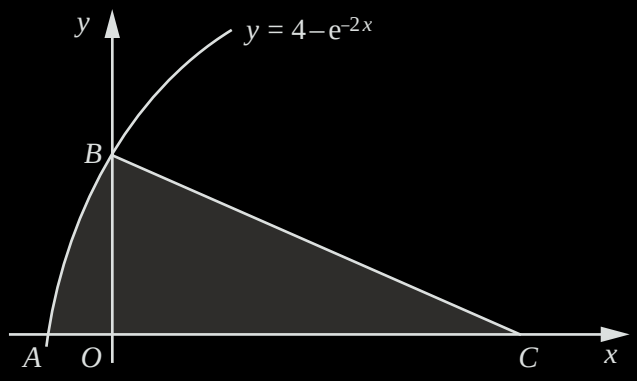
(b) Given that y is measured in radians, find the two smallest positive values of y such that

$$6\sin(2y + 1) + 5 = 0.$$ [5]

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12 Answer only **one** of the following two alternatives.

EITHER



The diagram shows part of the curve $y = 4 - e^{-2x}$ which crosses the axes at A and at B.

(i) Find the coordinates of A and of B. [2]

The normal to the curve at B meets the x-axis at C.

(ii) Find the coordinates of C. [4]

(iii) Show that the area of the shaded region is approximately 10.3 square units. [5]

OR

The variables x and y are related by the equation $y = 10^{-A}b^x$, where A and b are constants. The table below shows values of x and y.

x	15	20	25	30	35	40
y	0.15	0.38	0.95	2.32	5.90	14.80

(i) Draw a straight line graph of $\lg y$ against x, using a scale of 2 cm to represent 5 units on the x-axis and 2 cm to represent 0.5 units on the $\lg y$-axis. [2]

(ii) Use your graph to estimate the value of A and of b. [4]

(iii) Estimate the value of x when $y = 10$. [2]

(iv) On the same diagram, draw the line representing $y^5 = 10^{-x}$ and hence find the value of x for which $10^{A - \frac{x}{5}} = b^x$. [3]

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